

AUTOMATIC EMPATHY DETECTION

➤ THE TASK:

Detecting empathy from the natural language in different domains.

➤ THE APPROACH:

Employing multi-task training with knowledge distillation to incorporate knowledge from available resources (EMOTION and SENTIMENT).

- **Applications of analyzing and detecting empathy can be viewed from numerous perspectives:**
 - **medical and healthcare**
 - **human-computer interaction**
 - **neuroscience**
 - **philosophy**
 - **psychology**
 - **education**
- **Social platforms facilitate expressing empathy through natural language and text-based communication.**
- **Note that empathetic sentences are rich in emotion and sentiment. (When expressing empathy, emotional behavior is often displayed).**

- To analyze and understand empathy at scale, we must devise models to detect empathy from the natural language.
- Effectively training such models depends on the presence of quality labeled data.
- Existing datasets on the task of text-based empathy classification are small in size.
- To address the small data issue, we study the use of data-rich tasks related to empathy. Their correlation will be utilized in a multi-task learning setup.

➤ **The principle:**

- It is beneficial for various relevant tasks to be **learned jointly** so that each task can benefit from the knowledge learned in other tasks (Fukuda et al., 2017; Zagoruyko and Komodakis, 2016; Ma et al., 2018).
- **Multi-task training with knowledge distillation** technique is introduced in (Clark et al., 2019):
Kevin Clark, Minh-Thang Luong, Urvashi Khandelwal, Christopher D. Manning, and Quoc V. Le. 2019. Bam! Born-again multi-task networks for natural language understanding. *arXiv preprint arXiv: 1907.04829*.

Distilling Knowledge for Empathy Detection **(Hosseini and Caragea, 2021)**

- **Multi-task training with knowledge distillation is used (Hosseini and Caragea, 2021), for empathy detection, to incorporate knowledge into empathetic content from**
 - **emotion****and**
 - **sentiment.**

- **Two available resources are used as the external knowledge:**
 - **EmoNet (Abdul-Mageed and Ungar, 2017) – an emotion detection dataset**
 - **SST (Socher et al., 2013) – a sentiment classification dataset****(Note that EmoNet and SST are employed as single-task models to teach a multi-task model to detect empathy).**

RESULT:

- **Hosseini and Caragea (2021) show that the multi-task training with knowledge distillation outperforms strong baselines on two empathy datasets collected from different platforms on two different domains:**
 - **news – the dataset NewsEmp by Buechel et al. (2018)**
 - **health – the dataset TwittEmp, created from Twitter on health posts, by Hosseini and Caragea (2021).**
- ✓ **Note that a higher granularity of empathy is also explored in (Hosseini and Caragea, 2021)..**

Detecting empathy with multi-task learning

Hosseini and Caragea (2021) propose using multi-task learning with knowledge distillation and teacher annealing to leverage knowledge from available resources of sentiment and emotion for empathy detection.

- **Knowledge distillation (KD) is a prominent model compression technique for deep neural networks in which the knowledge of a trained large teacher model is transferred to a smaller student model.**

Multi-Task Learning - 1

- In multi-task learning (MTL) a target task is learned by employing knowledge from related auxiliary tasks. Knowledge learned in one task is shared across all tasks.
- Here:
 - the target task is empathy detection
 - the auxiliary tasks are emotion and sentiment classification.
- All the models are built on top of the pre-trained BERT language model.
- In MTL, the bottom layers (corresponding to BERT) are shared across all three tasks. The top layers are specific for each task (see Fig. 1, right side).
- A fully connected layer is used for each task, followed by softmax for classification.
- SOFTMAX
- Recall that logistic regression produces a decimal between 0 and 1.0. For example, a logistic regression output of 0.8 from an email classifier suggests an 80% chance of an email being spam and a 20% chance of it being not spam. Clearly, the sum of the probabilities of an email being either spam or not spam is 1.0.
- Softmax extends this idea into a multi-class world. That is, Softmax assigns decimal probabilities to each class in a multi-class problem. Those decimal probabilities must add up to 1.0. This additional constraint helps training converge more quickly than it otherwise would.
- Fully connected layers:
- Neural networks are a set of dependent non-linear functions. Each individual function consists of a neuron (or a perceptron). In fully connected layers, the neuron applies a linear transformation to the input vector through a weights matrix. A non-linear transformation is then applied to the product through a non-linear activation function f .

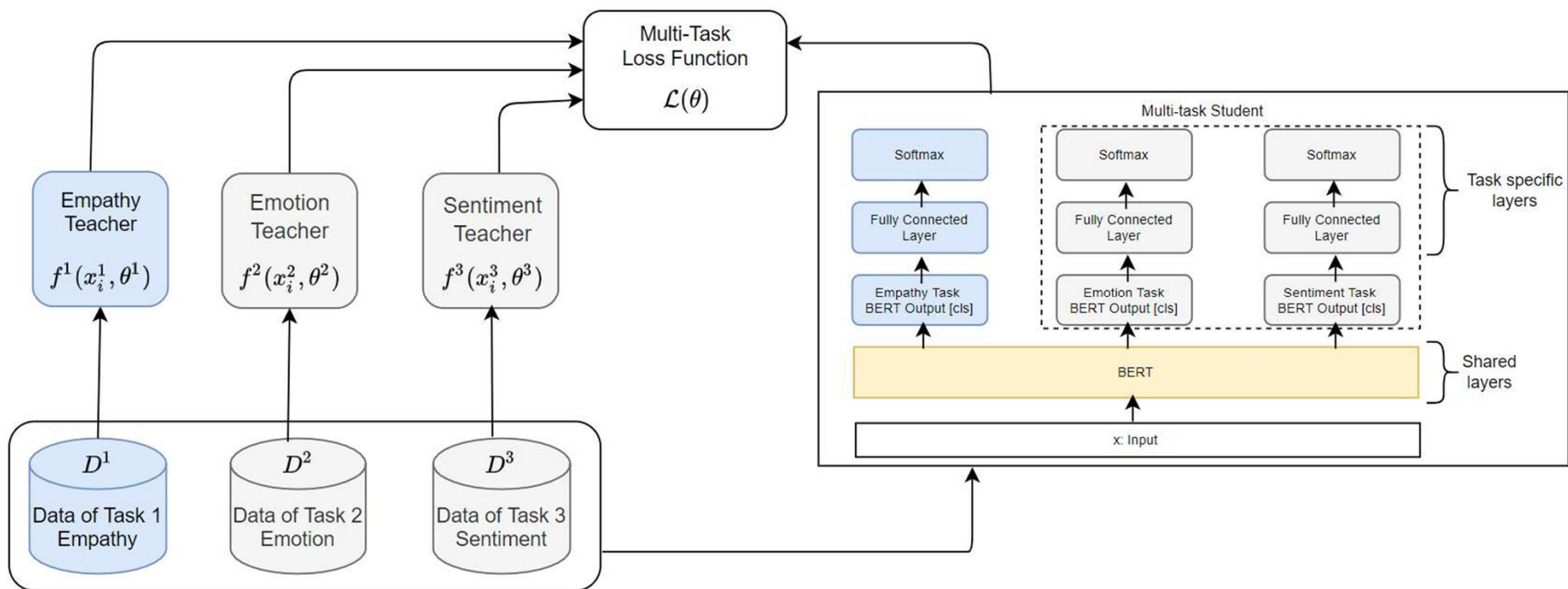


Figure 1: Multi-Task Training with Knowledge Distillation.

Multi-Task Learning – 2

- During MTL training, examples from the three tasks are shuffled together and the sum of the losses of all three tasks is minimized using backprop.

(Backpropagation or backprop is a widely used algorithm for training feedforward artificial neural networks, namely artificial neural networks in the case of which connections between the nodes do NOT form a cycle).

- That is:

Let $\mathcal{D}^\tau = \{(x_i^\tau, y_i^\tau)\}_i$ be the training set for task τ , where τ could be any of the three tasks (empathy, emotion, or sentiment).

The loss of the MTL model with parameter θ is:

$$\mathcal{L}(\theta) = \sum_{\tau=1}^3 \sum_{(x_i^\tau, y_i^\tau) \in \mathcal{D}^\tau} \ell(y_i^\tau, f^\tau(x_i^\tau, \theta)) \quad (1)$$

where $f^\tau(x_i^\tau, \theta)$ is the output of model θ on the input x_i^τ and ℓ is the cross-entropy loss.

➤ **Reminder:**

- **Cross-entropy is defined as**
$$L_{CE} = - \sum_{i=1}^n t_i \log(p_i),$$
 for n classes, where t_i is the truth label and p_i is the Softmax probability for the i^{th} class.
- **Cross-entropy loss is used when adjusting model weights during training. The aim is to minimize the loss. A perfect model has a cross-entropy loss of 0.**
- **Use the cross entropy loss function in both Tensorflow and PyTorch and log to Weights & Biases.**

Multi-Task Learning with Knowledge Distillation and Teacher Annealing – 1

- Hosseini and Caragea (2021) use the MTL model that distills knowledge from the auxiliary tasks into the target task proposed by (Clark et al., 2019).
- This model employs the idea of applying knowledge distillation with the purpose that single-task models (teachers) teach a multi-task model (student) so that the student becomes better than the teachers.
- This is distilling knowledge using a teacher-student framework, in which the student model learns the knowledge offered by the teachers' output.
- During training, as before, various tasks' examples are mixed jointly and the aggregated loss over all three tasks is minimized.

Multi-Task Learning with Knowledge Distillation and Teacher

Annealing – 2

- Formally, let $\mathcal{D}^\tau = \{(x_i^\tau, y_i^\tau)\}_i$ be the training set for task τ (empathy, emotion or sentiment).
- A single-task (teacher) model, denoted θ^τ , is trained on each task τ ($\tau = 1, 2, 3$), which produces output $f^\tau(x_i^\tau, \theta^\tau)$ on the input x_i^τ (see Fig. 1).
- Then, a multi-task shared (student) model with parameters θ (right side of Fig. 1) learns to imitate the output of the single-task (teacher) models θ^τ (left side of Fig. 1). The loss of the multi-task (student) model becomes:

$$\mathcal{L}(\theta) = \sum_{\tau=1}^3 \sum_{(x_i^\tau, y_i^\tau) \in \mathcal{D}^\tau} \ell\left(f^\tau(x_i^\tau, \theta^\tau), f^\tau(x_i^\tau, \theta)\right) \quad (2)$$

Multi-Task Learning with Knowledge Distillation and Teacher Annealing – 3

➤ **Note that:**

- **Formula (2) shows us that the MTL model with knowledge distillation is optimized based on teachers' predictions.**
- **However: emulating the teacher model in knowledge distillation may limit the student model to transcend the teacher model. That is why Clark et al. (2019) use a training strategy called teacher annealing.**

Multi-Task Learning with Knowledge Distillation and Teacher Annealing – 4

- The MTL with knowledge distillation and teacher annealing combines gold-standard with predictions:

$$\mathcal{L}(\theta) = \sum_{\tau=1}^3 \sum_{(x_i^{\tau}, y_i^{\tau}) \in \mathcal{D}^{\tau}} \ell\left(\lambda y_i^{\tau} + (1 - \lambda) f^{\tau}(x_i^{\tau}, \theta^{\tau}), f^{\tau}(x_i^{\tau}, \theta)\right) \quad (3)$$

where λ is linearly increased from 0 to 1 over the course of training.

- Remark: This approach benefits the student model to outperform its teachers.

DATA

- We incorporate external knowledge from the data-rich tasks of emotion and sentiment to detect empathy.
- We use:
 - SST-2 (Socher et al., 2013)
 - EmoNet (Abdul-Mageed and Ungar, 2017)
- We employ EmoNet and SST as single-task models to teach a multi-task model to detect empathy.
- We model empathy in a binary setting.

➤ **SST-2:**

- **binary dataset for sentiment analysis;**
- **consists of sentences from movie reviews and their sentiment (positive and negative);**
- **has 67,349 samples in the training set, 872 samples in the validation set, and 1,821 samples in the test set.**

➤ **EmoNet:**

- a dataset for emotion detection from Twitter;
- Hosseini and Caragea (2021) use the EmoNet version that contains tweets annotated with Plutchik-8 emotions, namely
 - ✓ joy
 - ✓ trust
 - ✓ fear
 - ✓ surprise
 - ✓ sadness
 - ✓ disgust
 - ✓ anger
 - ✓ anticipation
- has 41,669 samples in training, 5,166 samples in validation, and 5,214 samples in test.

Robert Plutchik (21 October 1927 – 29 April 2006) was a professor emeritus at the Albert Einstein College of Medicine. He was also a psychologist. His research interests included the study of emotions, the study of suicide and violence, and the study of the psychotherapy process.

Plutchik proposed a psychoevolutionary classification approach for general emotional responses. He considered there to be eight primary emotions anger, fear, sadness, disgust, surprise, anticipation, trust, and joy. Plutchik argues for the primacy of these emotions by showing each to be the trigger of behavior with high survival value.

DATASETS for modeling empathy on

- the **NewsEmp** dataset (Buechel et al., 2018)
- the **TwittEmp** dataset (Hosseini and Caragea, 2021)

NewsEmp

- contains 1,860 messages written in reaction to news articles rated with a numeric level of empathy and distress on a 7-point scale;
- Buechel et al. (2018) provided empathy binary labels, indicating if a message contains empathetic content or not;
- Hosseini and Caragea (2021) split the dataset into three sets of train, validation, and test with:
 - 80% of data used for training
 - 10% for validation
 - 10% for test

TwittEmp

- **dataset annotated by fine-grained empathy direction (seeking vs. providing);**
- **contains 3,000 English tweets;**
- **these tweets are annotated with three categories:**
 - ✓ **seeking-empathy**
 - ✓ **providing-empathy**
 - ✓ **none**
- **the collected data is related to the cancer topic;**
- **the Twitter streaming API, starting from July 2015 to August 2020, was used;**
- **requirement for selection: empathetic tweets must contain at least one of 200 high-rating empathy words, collected from the empathy and distress lexicon by Sedoc et al. (2019), plus the word “cancer”;**
- **as part of the preprocessing, duplicate tweets were removed and links and usernames were replaced with <URL> and <USER>, respectively.**

- **Modeling empathy (in a binary setting) in the TwittEmp dataset:**
 - keep tweets with labels *seeking-empathy* and *providing empathy* as positive samples;
 - keep tweets in the *none* class as negative samples;
 - split the dataset into three sets of train, validation and test with:
 - 80% of data for training
 - 10% of data for validation
 - 10% of data for test

Comparison

➤ average length of a tweet

- TwittEmp: around 37 words (max = 62 words); average number of 3 sentences per tweet;
- NewsEmp: 82 words (max = 163 words); average number of 5 sentences per message.

TESTS

Detecting Fine-grained Empathy

- **The goal:** given a tweet, classify it into one of the two categories
 - seeking-empathy
 - providing-empathy

- **Action:**
 1. create two classifiers (in a binary setting), one to detect tweets seeking empathy and one to detect tweets providing empathy.
 - ✓ **The seeking-classifier:**
 - keep *seeking-empathy* as positive samples;
 - combine the two classes of *none* and *providing-empathy* as negative samples.
 - ✓ **The providing-classifier:**
 - keep *providing-empathy* as positive samples
 - combine the two classes of *none* and *seeking-empathy* as negative samples.

 2. split the datasets:
 - ✓ 60% of data for training
 - ✓ 20% for validation
 - ✓ 20% for testing

Models

➤ Hosseini and Caragea (2021) contrast

- **the multi-task learning with knowledge distillation and teacher annealing that learns from teachers' outputs and one-hot labels (Eq. 3)**

with

- **the multi-task learning that uses one-hot labels (Eq. 1)**

and with the following baselines:

- **standard neural methods (CNN, LSTM, ConvLSTM, BiLSTM)**
- **pre-trained language models (fine-tuning BERT)**

Test Results

See the organized tests and test results in:

Mahshid Hosseini and Cornelia Caragea, Distilling Knowledge for Empathy Detection. EMNLP 2021, November 7-11, pp. 3713-3724.

General Conclusion

- **A multi-task training with knowledge distillation technique has been used to incorporate knowledge into empathetic content from EmoNet and SST. This approach achieves better results on two datasets from different domains.**
- **Distilling knowledge from available related resources on emotion and sentiment can be effectively used to inform empathy classification.**